

Look! A planet

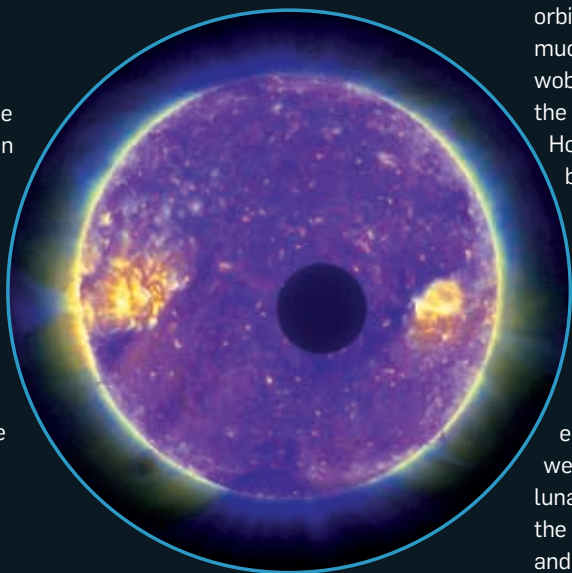
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Little do we know about the vast expanse above us. Each time we look above, it's an entirely new scenario. Some elements, like the constellations, remain the same, however light of heavenly bodies from parsecs away continue to reach us as we go about our lives. It brings with it the history of the planet. How it formed and what changes have taken place on its surface are all contained in this cosmic beam of light. Here's how we go about interpreting this data to detect new planets in the universe.

Methods to find new planets

Till the late 90's, astronomers weren't confident about the existence of extra-solar planets, i.e. planets outside our solar system. Multiple theories existed but no distinct method was devised to find them. Now, we're coming across new planets on a daily basis using the following methods.



Wobbling stars

When a planet orbits a star, there's mutual gravitational attraction. This gravity tugs the star in the plane of the planet's orbit. This attributes to a slight wobble for the star. With each subsequent planet

orbiting the body, the wobble becomes much more complex. By studying this wobbling of a star, astronomers figure out the number of planets orbiting the star.

How we figure out whether a star is wobbling or not is by observing the planet's red/blue shifting.

Eclipses

An eclipse as we all know happens when three or more planets line up and light from one body is blocked by another. We have solar and lunar eclipses happening every year. In 2011, we have four partial solar and two total lunar eclipses happening. Watch out for the solar eclipse on the 25th of this month and a lunar eclipse on 10th of December. Similarly, extrasolar planets also eclipse their stars. When we observe these distant stars, there's a slight reduction of the star's brightness when its planet passes between the star and earth. This tells us that a planet is indeed orbiting the star and the time period between subsequent eclipses gives the time that planet takes to circle its star, which gives us an idea about the planet's proximity with the star.

Looking straight at it

One of the simple sounding but much difficult methods is to peek straight at the region where a planet is suspected to be. Light comes in a very wide spectrum and when observed in the visible range, the star's brightness will be much more than the planet's. This will practically blind the telescope and the planet won't be seen. However, when the infrared region of the light spectrum is observed, the relative brightness of the planet increases as it emits its own infrared light as well. Thus, letting us know of its presence.

Polarimeters

Prior to being reflected off an object, light is non-polarised and when it bounces off a planet it becomes polarised. We have pola-

Goldilocks zone

Our planet is the only one in the solar system to support life. Though there are multiple parameters which need to be fulfilled in order for this to happen, one particular parameter is the distance of the planet from the sun. Any planet that is too close to the sun like Mercury or Venus is too hot to sustain life whereas planets far from the star like Saturn and Neptune are too cold to sustain life. According to the size of the star, this narrow band called the Goldilocks zone has been proven favourable enough to sustain life. Earlier, even Earth wasn't in the Goldilocks zone but after finding life in the most unfavourable places on the surface, astronomers have widened this band. In our solar system, Earth and Mars are the only two planets that are can of sustaining life. So when we detect a new planet, one of the first things we do is check whether the new planet falls within the Goldilocks zone for that particular star.

